

2.8 Solution: (a) Assume that the two line are located at $(+c, 0)$ and $(-c, 0)$, $c > 0$, with linear charge densities λ and $-\lambda$, respectively. Then, the potential at point (x, y) is given by

$$\Phi(x, y) = \frac{\lambda}{4\pi\epsilon_0} \log \left(\frac{(x+c)^2 + y^2}{(x-c)^2 + y^2} \right).$$

Therefore, the equipotential surface can be determined by the condition $\Phi(x, y) = V$, or

$$\frac{(x+c)^2 + y^2}{(x-c)^2 + y^2} = A,$$

where $A = \exp(4\pi\epsilon_0 V/\lambda)$. This leads to the following equation,

$$(1-A)x^2 + 2(1+A)cx + (1-A)c^2 + (1-A)y^2 = 0,$$

or equivalently,

$$\left[x - \frac{A+1}{A-1}c \right]^2 + y^2 = \frac{4A}{(A-1)^2}c^2.$$

Thus, the equipotential surfaces are circular cylinders, with centers at

$$\left(-\frac{A+1}{A-1}c, 0 \right)$$

and radii

$$R = \frac{2\sqrt{A}}{|A-1|}.$$

Also, the equipotential surface reduces to the yz -plane when $A = 1$ or $V = 0$.

(b) For a cylinder with radius a , we can write

$$a^2 = \frac{4A}{(A-1)^2}c^2.$$

Assume the center of this cylinder lies on the positive x -axis, then $A > 1$, and

$$A_a = \frac{\sqrt{c^2 + a^2} + c}{\sqrt{c^2 + a^2} - c},$$

and

$$V_a = \frac{\lambda}{4\pi\epsilon_0} \log \left(\frac{\sqrt{c^2 + a^2} + c}{\sqrt{c^2 + a^2} - c} \right).$$

The center of the cylinder is located at

$$x_a = \frac{A_a + 1}{A_a - 1}c = \sqrt{c^2 + a^2}.$$

Similarly, for a cylinder with radius b whose center is on the negative x -axis, we have

$$A_b = \frac{\sqrt{c^2 + b^2} - c}{\sqrt{c^2 + b^2} + c}, \quad V_b = \frac{\lambda}{4\pi\epsilon_0} \log \left(\frac{\sqrt{c^2 + b^2} - c}{\sqrt{c^2 + b^2} + c} \right),$$

and

$$x_b = -\sqrt{c^2 + b^2}.$$

The potential difference between the two cylinders is

$$\Delta V = V_a - V_b = \frac{\lambda}{4\pi\epsilon_0} \log \left(\frac{(\sqrt{c^2 + a^2} + c)(\sqrt{c^2 + b^2} + c)}{(\sqrt{c^2 + a^2} - c)(\sqrt{c^2 + b^2} - c)} \right) = \frac{\lambda}{2\pi\epsilon_0} \log \left(\frac{(\sqrt{c^2 + a^2} + c)(\sqrt{c^2 + b^2} + c)}{ab} \right).$$

Since the cylinders are separated by a distance of d , we must have

$$\sqrt{c^2 + a^2} + \sqrt{c^2 + b^2} = d. \quad (1)$$

Therefore,

$$(\sqrt{c^2 + a^2} + c)(\sqrt{c^2 + b^2} + c) = \sqrt{c^2 + a^2}\sqrt{c^2 + b^2} + cd + c^2.$$

Meanwhile,

$$d^2 = (\sqrt{c^2 + a^2} + \sqrt{c^2 + b^2})^2 = a^2 + b^2 + 2c^2 + 2\sqrt{c^2 + a^2}\sqrt{c^2 + b^2}.$$

Then,

$$\sqrt{c^2 + a^2}\sqrt{c^2 + b^2} = \frac{d^2 - a^2 - b^2}{2} - c^2.$$

Also from Eq. (1), we have

$$c^2 + a^2 = d^2 + c^2 + b^2 - 2d\sqrt{c^2 + b^2},$$

and

$$\sqrt{c^2 + b^2} = \frac{d^2 - a^2 + b^2}{2d}.$$

Finally, we have

$$c^2 = (c^2 + b^2) - b^2 = \left(\frac{d^2 - a^2 + b^2}{2d} \right)^2 - b^2 = \frac{(d^2 - a^2 - b^2)^2}{4d^2} - \frac{a^2 b^2}{d^2},$$

and

$$cd = \left[\frac{1}{4}(d^2 - a^2 - b^2)^2 - a^2 b^2 \right]^{1/2}.$$

Putting everything together, we will get

$$(\sqrt{c^2 + a^2} + c)(\sqrt{c^2 + b^2} + c) = ab \left\{ \frac{d^2 - a^2 - b^2}{2ab} + \left[\left(\frac{d^2 - a^2 - b^2}{2ab} \right)^2 - 1 \right]^{1/2} \right\},$$

and

$$\Delta V = \frac{\lambda}{2\pi\epsilon_0} \log \left(\frac{d^2 - a^2 - b^2}{2ab} + \sqrt{\left(\frac{d^2 - a^2 - b^2}{2ab} \right)^2 - 1} \right).$$

Since

$$\operatorname{arccosh} x = \log(x + \sqrt{x^2 - 1}),$$

we can express the potential difference as

$$\Delta V = \frac{\lambda}{2\pi\epsilon_0} \operatorname{arccosh} \left(\frac{d^2 - a^2 - b^2}{2ab} \right),$$

and the capacitance becomes

$$C = \frac{\lambda}{\Delta V} = \frac{2\pi\epsilon_0}{\operatorname{arccosh}\left(\frac{d^2 - a^2 - b^2}{2ab}\right)}.$$

(c) For $a, b \ll d$, we can expand the the arccosh function as

$$\operatorname{arccosh}\left(\frac{d^2 - a^2 - b^2}{2ab}\right) = \log\left(\frac{d^2}{ab}\right) - \left(\frac{a}{d}\right)^2 - \left(\frac{b}{d}\right)^2.$$

To the leading order,

$$C = \frac{2\pi\epsilon_0}{\log\left(\frac{d^2}{ab}\right)},$$

which agrees with Prob. 1.7.

(d) Similar to part (b), the cylinder of radius b has its center on the positive x -axis, and we must have

$$A_b = \frac{\sqrt{c^2 + b^2} + c}{\sqrt{c^2 + b^2} - c},$$

$$V_b = \frac{\lambda}{4\pi\epsilon_0} \log\left(\frac{\sqrt{c^2 + b^2} + c}{\sqrt{c^2 + b^2} - c}\right),$$

and the center

$$x_b = \frac{A_b + 1}{A_b - 1}c = \sqrt{c^2 + b^2}.$$

Now, the separation becomes

$$\sqrt{c^2 + a^2} - \sqrt{c^2 + b^2} = d,$$

assuming $a > b$. We can repeat the calculation as in part (b), which will give the capacitance as

$$C = \frac{2\pi\epsilon_0}{\operatorname{arccosh}\left(\frac{a^2 + b^2 - d^2}{2ab}\right)}.$$

For more details, check the scanned solution.

For $d = 0$, we have

$$\operatorname{arccosh}\left(\frac{a^2 + b^2 - d^2}{2ab}\right) = \log\left(\frac{a}{b}\right),$$

and we will recover the well known result for concentric cylinders,

$$C = \frac{2\pi\epsilon_0}{\log\left(\frac{a}{b}\right)}.$$